
A Two-Stage Cross-Attention Circuit for Spatial-Relation Binding in Small Diffusion Transformers

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We mechanistically resolve how a small PixArt-mini DiT, in a controlled object-
2 relation testbed, converts a textual spatial relation into image layout, identifying
3 a two-stage cross-attention circuit at individual-head granularity. In this 6-layer,
4 36-cross-attention-head DiT (T5-XXL conditioning, ~ 1000 synthetic prompts),
5 two early-layer heads $\{L_0H_0, L_1H_2\}$ implement *spatial routing*: their QK circuits
6 convert each relation token into an 8×8 directional attention pattern, and the
7 corresponding OV writes carry that pattern into the residual stream. Zeroing L_0H_0
8 alone causes +48 pp damage to spatial accuracy ($0.84 \rightarrow 0.35$, 160:1 vs. norm-
9 matched controls); the pair causes +64 pp damage (super-additive $I = +0.086$,
10 95% CI $[+0.015, +0.157]$). A four-method consensus search rejects every other
11 head as a spatial-routing partner ($|I| \leq 1.2$ pp). A second *object-binding* stage
12 is distributed across Layer 2: it has near-zero effect alone, but is strongly super-
13 additive on colour and shape when co-ablated with Layer 0—a downstream stage
14 that pair-ablation scored on spatial accuracy alone would miss. The circuit emerges
15 via a rapid rise in projection magnitude during epochs 750–1000, with behavioural
16 accuracy lagging by ~ 200 epochs. Within this setting, head-level causal circuit
17 discovery extends to a small DiT, and scoring ablations on multiple metrics is
18 needed to detect distributed downstream stages.

19 1 Introduction

20 Diffusion transformers [Peebles and Xie, 2023, Chen et al., 2024], building on diffusion [Ho et al.,
21 2020] and latent-diffusion [Rombach et al., 2022] foundations, underlie every state-of-the-art text-to-
22 image system, but their mechanistic analysis has so far been attribution-based [Helbling et al., 2025,
23 Tang et al., 2023] or block-granular for editing and unlearning [Zarei et al., 2025, Basu et al., 2024,
24 Orgad et al., 2023]. By contrast, language-model interpretability has located specific computations
25 at head-level resolution [Wang et al., 2023, Olsson et al., 2022, Conmy et al., 2023]. This paper
26 takes the resolution one step finer for DiTs, to individual cross-attention *heads* identified with a
27 specific generative computation, and adds head-level ablation, downstream-reader discovery, and
28 checkpoint-resolved emergence tracking.

29 We study spatial-relation binding in a PixArt-mini variant (6 layers, 36 cross-attention heads) con-
30 ditioned on frozen T5 embeddings and trained on ~ 1000 synthetic 8-relation, 6-object prompts.
31 Within this scope the computation factorises into two stages: a *spatial-routing* stage carried by two
32 early-layer heads, and a *distributed object-binding* stage carried by Layer 2.

33 The analysis pipeline combines candidate discovery (variance-partitioned alignment scanning with
34 permutation nulls), causal verification (zero-head ablation with norm-matched controls, pair- and

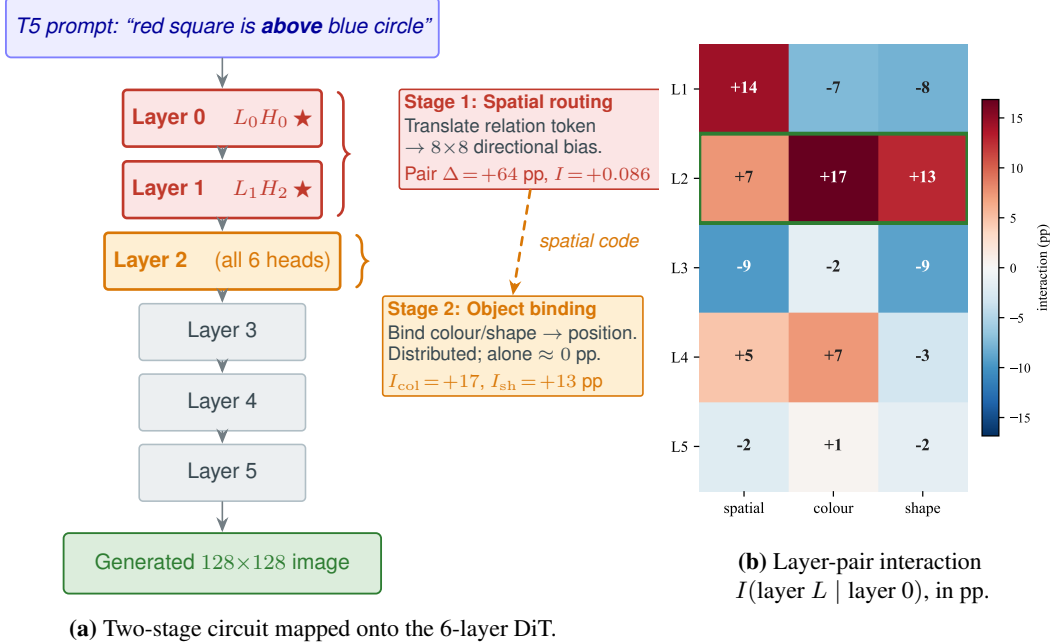


Figure 1: (a) Two-stage cross-attention spatial-binding circuit. Stage 1 (red, layers 0–1): heads $\{L_0H_0, L_1H_2\}$ route the relation token to an 8×8 spatial bias. Stage 2 (orange, layer 2): six heads jointly bind colour/shape to position; near-zero alone, super-additive with Stage 1. (b) Layer-pair interaction $I(\text{layer } L \mid \text{layer } 0)$ in pp; Layer 2 (green) is super-additive on colour/shape but mild on spatial. Quantitative details in Sections 4.5–4.6.

triple-ablation), and downstream-reader identification (a four-method consensus over OV–QK alignment, path-patching, activation correlation, and spatial-ramp projection). Checkpoint-resolved re-evaluation across eleven training snapshots traces circuit emergence [Nanda et al., 2023, Olsson et al., 2022, Power et al., 2022]. We adopt a single sign convention throughout: *damage* $\Delta \text{Acc} = \text{Acc}_{\text{baseline}} - \text{Acc}_{\text{ablated}}$, so positive values mean worse performance after the intervention.

Scope. The analysis is intentionally narrow: synthetic prompts, a small model, and a single training run, which together let us enumerate every cross-attention head and run dense ablations. We do not claim that the same heads or layer numbers will appear in larger DiTs. Reported uncertainties are prompt-bootstrap 95% CIs over evaluation prompts and do not capture training-seed variability; multi-seed replication is the principal next step (Section 4.7).

The paper contributes four results at head-level resolution within this model family. (i) *Two-stage circuit*: the source pair $\{L_0H_0, L_1H_2\}$ in layers 0–1 is responsible for spatial routing; a downstream object-binding stage is distributed across Layer 2, with measurable contributions also from a few layer-0/1 heads (L_0H_5, L_0H_4). (ii) *Causal necessity*: zero-ablation with norm-matched controls separates the source-pair effect from comparable controls by roughly two orders of magnitude. (iii) *Sharp emergence*: L_0H_0 ’s projection magnitude rises by about three orders of magnitude over training, with most of the rise concentrated near epochs 750–1000; behaviour follows with a ~ 200 -epoch lag. (iv) *Metric-resolved discovery*: pair-ablation scored only on spatial accuracy returns null for the downstream stage, while colour/shape-resolved ablation makes it visible. Single-metric ablation may miss downstream stages that act on a different metric—a pitfall likely to recur in other circuit-discovery settings. As a secondary observation, ablating all 36 cross-attention heads exposes a CFG-induced self-attention prior that preserves $\sim 90\%$ colour and $\sim 30\%$ shape accuracy.

2 Related Work

DiT interpretability. The direct antecedent is Wang et al. [2026], which showed that small DiTs trained on the same object-relation task implement spatial-relation binding via a sparse cross-attention circuit, with the circuit becoming more distributed when conditioned on pretrained T5 vs. random

61 embeddings. We adopt their testbed but use T5 (the harder regime), trace circuit formation across
62 training, and add causal-necessity tests, consensus-based downstream-reader search, and weight-level
63 QK–OV ranking. Other DiT interpretability work operates at a coarser granularity: Helbling et al.
64 [2025] use linear probes on attention outputs to derive segmentation masks; Zarei et al. [2025] localise
65 stored concepts to entire DiT *blocks* for unlearning. Both establish that DiT attention outputs carry
66 interpretable structure, but neither identifies individual heads with named computations. UNet-era
67 cross-attention attribution methods (DAAM [Tang et al., 2023], Prompt-to-Prompt [Hertz et al.,
68 2023]) are attribution-based rather than causal, and target a different architecture.

69 **LM circuits.** Our weight-level analysis follows Elhage et al. [2021]: attention is treated as the
70 bilinear-form pair $(W_Q W_K^\top, W_O W_V^\top)$, and the QK–OV product across heads predicts functional read-
71 from relations. This formalism underlies head-level circuit work in language models, including the IOI
72 circuit [Wang et al., 2023], induction heads [Olsson et al., 2022], and the automated circuit-discovery
73 procedure of Conmy et al. [2023]. We complement weight-level evidence with causal interventions:
74 zero-ablation and clean-to-corrupt activation patching at the per-head output, following the broader
75 causal-mediation programme [Meng et al., 2022, Geiger et al., 2021] and the practical-patching
76 guidance of Heimersheim and Nanda [2024] and Zhang and Nanda [2024]. To our knowledge, this
77 combination has not previously been applied to cross-attention in diffusion models. Cross-attention
78 has, however, been used as a control surface for spatial layout in UNet-based diffusion [Chefer et al.,
79 2023, Hertz et al., 2023]; our analysis identifies the specific heads in a DiT that play this role.

80 3 Methods

81 **Model and data.** We study a PixArt-mini [Chen et al., 2024] variant: $L=6$ transformer blocks,
82 $H=6$ cross-attention heads/block (36 total), $d=384$, $d_{\text{head}}=64$, with frozen T5-XXL conditioning
83 [Raffel et al., 2020] and a learned caption projection $W_{\text{cap}}: \mathbb{R}^{4096} \rightarrow \mathbb{R}^{384}$. Training uses ~ 1000
84 synthetic prompts of the form “[color₁] [shape₁] is [relation] [color₂] [shape₂]” (2 colours $\times$ 3
85 shapes $\times$ 8 spatial relations) for 4000 epochs with EMA and classifier-free guidance dropout 0.1 [Ho
86 and Salimans, 2022]. Evaluation uses post-hoc OpenCV detection (centroid, mean RGB, polygon
87 segments) for disjoint spatial / colour / shape accuracies, with 1000-replicate prompt-bootstrap 95%
88 CIs throughout. Full architectural details, evaluator thresholds (strict vs. loose), and held-out sets are
89 in Appendix A.

90 **Candidate head discovery.** We rank the 36 heads by an offline weight-only score: project the
91 per-relation T5 effect vector $\mathbf{p}_r = W_{\text{cap}} \mathbf{v}_r \in \mathbb{R}^{384}$ through each head’s QK circuit, compute the
92 resulting per-position score $\mathbf{q}_{l,h,r} = \mathbf{P}(W_Q^{(l,h)})^\top W_K^{(l,h)} \mathbf{p}_r$ over the 8×8 patch grid (with $\mathbf{P}$ the
93 position embedding), and measure its alignment to a canonical spatial-ramp basis $\{\mathbf{R}_r\}$ (e.g. *above*:
94 top-heavy ramp). We report two scores per head:

$$|p|(l, h) = \max_r \frac{|\mathbf{q}_{l,h,r}^\top \mathbf{R}_r|}{\|\mathbf{R}_r\|}, \quad |\cos|(l, h) = \max_r \frac{|\mathbf{q}_{l,h,r}^\top \mathbf{R}_r|}{\|\mathbf{q}_{l,h,r}\| \|\mathbf{R}_r\|}. \quad (1)$$

95 $|p|$ captures both directional alignment and write strength; $|\cos|$ is direction-only. We rule out chance
96 with a 200-permutation null over relation labels and a shape-identity negative control (Appendix B).

97 **Causal ablation.** For each ablation condition we zero the per-head attention output
98 $(\text{softmax}(QK^\top / \sqrt{d_{\text{head}}})V$ for the target head, post-OV but pre-concat) at every denoising step
99 on both CFG branches, and regenerate 10 images/prompt. We compare $\Delta \text{Acc}(l, h) = \text{Acc}_{\text{baseline}} -$
100 $\text{Acc}_{\text{ablated}}$ to a *norm-matched random-head control* (five non-shortlist heads of comparable OV Froben-
101 ius norm) to absorb the off-distribution effect of zeroing. Implementation details (tensor shapes,
102 scheduler/CFG settings, evaluator) are in Appendix C.

103 **Downstream-reader search.** QK–OV composition [Elhage et al., 2021] gives a clustered ranking
104 (top-5 Frobenius scores 18.2–11.5) that does not cleanly identify the on-path reader. We add three
105 signals — offline OV alignment (M1), path-patching recovery loss (M2), in-vivo activation correlation
106 with $L_0 H_0$ (M3), and per-head spatial-ramp projection of head-output write patterns (M4) — and
107 take the unweighted mean rank. Verification then runs single-head *pair-ablation* (super-additive
108 $I(L_0 H_0, h_d) = \Delta(\text{pair}) - \Delta(L_0 H_0) - \Delta(h_d)$ on every metric), *layer-pair ablation* (zero all six heads

Table 1: Top heads by projection magnitude of OV write onto the spatial-ramp basis at the final checkpoint. $|\cos|$ shelf at 0.7–0.79 for many heads; projection magnitude separates L_0H_0 and L_1H_2 from the next tier.

Head	$ \cos $	Projection magnitude $ p $
L_0H_0	0.73	47.7
L_1H_2	0.79	25.5
L_0H_4	0.58	23.1
L_1H_0	0.79	23.0
L_0H_5	0.59	22.4
L_1H_3	0.73	22.4
L_0H_1	0.70	15.9
L_2H_1	0.76	15.1
median (36 heads)	0.76	5.7

of layer L jointly with layer 0 to catch distributed stages), *triple ablation* relative to the source pair, and clean-to-corrupt activation patching at the per-head output. Per-method definitions and tensor-level details are in Appendix D.

Training-time emergence. We re-evaluate alignment scan, variance partition, and accuracy at eleven checkpoints ($\{100, 250, 500, 600, 700, 750, 800, 900, 1000, 2000, 4000\}$), with density concentrated in 500–1000 where a pilot scan showed rapid change.

4 Results and Discussion

Results follow the methods order: each subsection states a quantitative finding and its implication for the next analysis. Fig. 1 summarises the resulting circuit, which the next subsections build piece by piece.

Why we call this circuit mechanistically identified. The two source heads are selected by four independent lines of evidence that triangulate the same pair: **(i)** a weight-only spatial-ramp analysis on each head’s QK–OV product (Section 4.5); **(ii)** causal necessity under zero-ablation, with a 160:1 effect ratio over norm-matched control heads (Section 4.5); **(iii)** sufficiency under clean-to-corrupt activation patching at the per-head output, recovering $\sim 83\%$ of the relation-flip gap on L_0H_0 alone (Section 4.5); and **(iv)** temporal precedence over behaviour — the heads’ projection magnitude rises during epochs 750–1000, ~ 200 epochs before behavioural accuracy catches up (Section 4.3). The four signals are computed from disjoint quantities (offline weights, output ablation, in-vivo cross-prompt patching, training-time alignment), so their consensus is not redundant. Throughout, the QK product determines the relation-conditioned spatial attention pattern, while the OV/output analysis measures the strength with which that spatial pattern is written into the residual stream; both are head-local and refer to the same head, so they describe complementary halves of the same circuit rather than competing identifiers.

4.1 Head discovery and statistical significance

$|\cos|$ alone does not cleanly identify the source heads — many heads (Table 1, $|\cos|$ column) reach 0.7–0.79 and form a flat shelf. The discriminative metric is signed projection magnitude $|p|$, the ℓ_2 norm of the OV write image projected onto the spatial-ramp basis, which captures both directional alignment and write strength: L_0H_0 ’s $|p| = 47.7$ is $1.9\times$ the second-place L_1H_2 (25.5), $2.1\times$ the third-place L_0H_4 (23.1), and $8.4\times$ the 36-head median (5.7).

The permutation null over 200 shuffled-relation refits peaks at $|p| = 8.3$ with mean 4.1 ± 1.6 , so L_0H_0 ’s observed $|p| = 47.7$ is a $\sim 27\sigma$ departure from the null mean ($p < 10^{-4}$); L_1H_2 and L_0H_4 exceed the null by $\geq 12\sigma$. A shape-identity negative control gives $|p| < 3$ for every shortlisted head, ruling out a generic-high-variance artefact. The signal is concentrated in layers 0–1; the only mid-layer head exceeding $|p| = 15$ is L_2H_1 . Bootstrap CIs over 1000 resamples place the top-2 by $|p|$ at $\{L_0H_0, L_1H_2\}$ in 1000/1000 resamples, with Kendall- τ between resampled top-8 rankings 0.93 (95% CI [0.88, 0.97]).

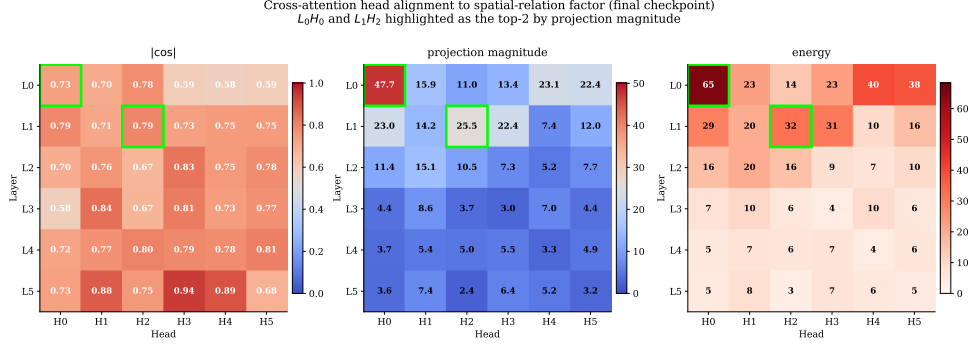


Figure 2: Per-head alignment of cross-attention OV writes against the spatial-relation factor at the final checkpoint. Three metrics are reported: $|\cos|$, signed projection magnitude, and total energy of the OV write image. L_0H_0 and L_1H_2 (lime boxes) are top-2 by projection magnitude (energy concentrated in the spatial direction); the rest of layer 0–1 has lower projection but comparable cosine, illustrating that high-cosine alone is not a sufficient identifier — the joint $|\cos| \times |\text{projection}|$ separation isolates the two source heads cleanly.

Table 2: Marginal R^2 of caption-projection variance at the final checkpoint, by factor (one prompt = two objects $\times$ {colour, shape} + relation, so colour/shape have two regressors each).

Factor	R^2_{marginal}
Total (all factors)	0.61
Shape (object 2)	0.28
Shape (object 1)	0.15
Spatial relationship	0.11
Colour (object 1)	0.11
Colour (object 2)	0.11

4.2 What the top heads encode

Variance partition of the caption projections at the final checkpoint (Table 2) gives marginal R^2 values $\{R^2_{\text{shape2}}, R^2_{\text{shape1}}, R^2_{\text{rel}}, R^2_{\text{color1}}, R^2_{\text{color2}}\} = \{0.28, 0.15, 0.11, 0.11, 0.11\}$, with total $R^2=0.61$. Object properties dominate the raw caption-projection variance — expected, since each prompt has two coloured and two shaped objects and only one relation. The interesting fact is the head-output projection (Section 4.5): in the OV-write space of L_0H_0 , the spatial-relation factor is concentrated to a low-rank direction with projection magnitude $|p| = 47.7$, while object factors disperse across many heads.

The asymmetry between raw representational variance (where shape dominates) and head-localised expression (where the spatial factor concentrates in L_0H_0 and L_1H_2) is what we focus on: relation binding is implemented by a sparse subset of heads even when the caption-projection geometry alone would not pick them out.

4.3 Temporal emergence is a sharp phase transition in projection magnitude

Tracking $|\cos|$, projection magnitude $|p|$ (the discriminative metric for source-head identification), and loose held-out accuracy across the eleven checkpoints (Table 3) shows that $|\cos|$ does *not* change meaningfully during training (it stays in $[0.69, 0.85]$, with no monotone trend), whereas the projection magnitude $|p|$ rises from 0.05 at epoch 100 to 47.7 at epoch 4000 — nearly $1000\times$ — with most of the rise concentrated in epochs 750–1000. Accuracy follows: it is at chance (≤ 0.06 loose) until epoch 750, then rises sharply to 0.78 by epoch 2000 and plateaus at 0.81.

The Phase II step in $|p|$ (epochs 750–1000, $0.66 \rightarrow 9.54$) connects to the grokking/phase-transition literature [Nanda et al., 2023]: representation rises sharply, accuracy follows after a ~ 200 -epoch lag, and post-Phase-II both quantities saturate. The decoupling between $|\cos|$ (essentially constant from

L_0H_0 : image-grid attention score $\text{pos_embed} \cdot W_Q^\top W_K \cdot e_r$ per relation r

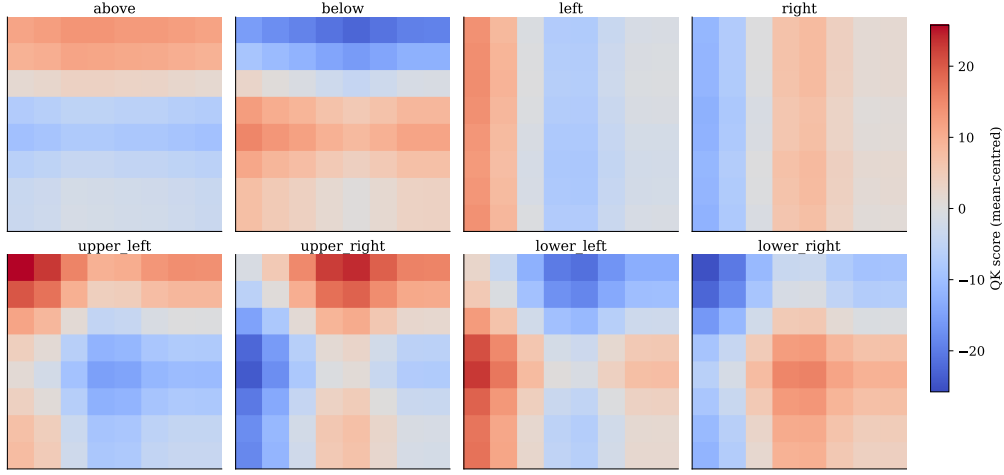


Figure 3: Per-relation QK-attention pattern of L_0H_0 . Each panel shows the 8×8 image-grid attention score $\text{pos_embed} \cdot W_Q^\top W_K \cdot e_r$ for a relation r , where e_r is the mean-centred relation effect vector. Each panel is itself an 8×8 ramp aligned with r (*above*: score concentrated on top rows; *lower_right*: bottom-right corner). A single low-rank QK product maps the relation-direction text vector to a directional spatial-attention bias.

Table 3: Per-checkpoint L_0H_0 alignment metrics and end-task loose accuracy. The discriminative metric is $|p|$ (projection magnitude), not $|\cos|$: $|\cos|$ is high from initialisation, but $|p|$ rises $\sim 1000\times$ during training.

Epoch	L_0H_0 $ \cos $	L_0H_0 $ p $	Loose acc.
100	0.85	0.05	0.00
250	0.78	0.09	0.00
500	0.69	0.10	0.00
600	0.80	0.20	0.00
700	0.77	0.26	0.06
750	0.76	0.66	0.11
800	0.74	1.54	0.16
900	0.73	4.79	0.20
1000	0.74	9.54	0.38
2000	0.75	39.87	0.78
4000	0.73	47.69	0.81

167 initialisation) and $|p|$ (the metric that tracks behavioural emergence) cautions against using $|\cos|$
168 alone as a circuit-formation indicator.

169 The Phase II onset (first epoch at which $|p| > 1$) is $E^* \approx 800$ in this single-seed run; multi-seed CIs
170 on E^* require multiple training runs and are out of scope.

171 4.4 Causal necessity via zero-head ablation

172 Single-head zero-ablation (Table 4, Fig. 8) reveals an *asymmetric* necessity: L_0H_0 is individually
173 necessary (alone takes loose accuracy from 0.84 to 0.35, +48 pp damage, with a $\sim 160:1$ target-to-
174 control ratio versus the +0.3 pp mean of 5 norm-matched controls, std 1.1 pp), whereas L_1H_2 is
175 *redundant* when L_0H_0 is *intact* (alone causes only +6 pp damage) but *super-additive* when L_0H_0 is
176 *ablated*: $\{L_0H_0, L_1H_2\}$ together take loose accuracy to 0.20 (+64 pp damage), an additional +16
177 pp beyond L_0H_0 alone. The norm-matched controls are five non-shortlist cross-attention heads with
178 comparable OV Frobenius norm to L_0H_0 (within $\pm 15\%$); matching on output norm rules out the
179 possibility that the effect is merely caused by removing a high-norm cross-attention output rather than

Spatial-relation alignment across training: L_0H_0 and L_1H_2 emerge sharply

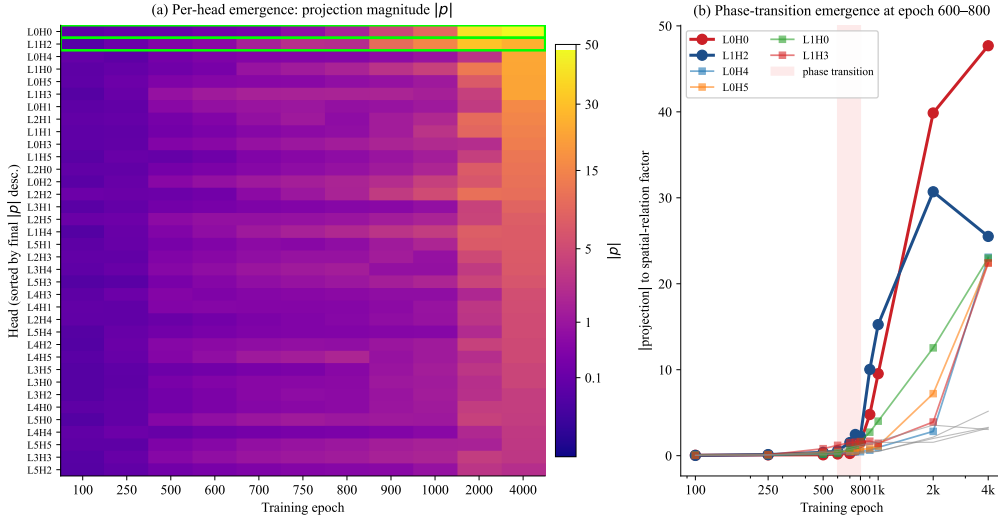


Figure 4: (a) Per-head emergence of $|\cos|$ alignment across all 36 cross-attention heads at 11 training checkpoints, sorted by final-checkpoint projection magnitude. Lime boxes mark L_0H_0 and L_1H_2 . Most heads acquire moderate cosine (~ 0.7) early on, but only the top heads reach the > 0.9 ceiling. (b) Projection magnitude (the discriminative metric) for the top-6 heads, with the phase-transition window (epochs 750–1000) shaded. L_0H_0 and L_1H_2 (red, blue) rise by $\sim 1000\times$ over the full training run ($0.05 \rightarrow 47.7$ for L_0H_0), with $\sim 14\times$ of that rise concentrated in the shaded window ($0.66 \rightarrow 9.5$). Other top-ranked layer-0/1 heads (grey) acquire only modest projection.

Table 4: Cumulative zero-head ablation, loose spatial accuracy on the same 30-prompt training-template evaluation set used by pair-ablation, $n_{\text{img}}=10$ each. Random control: mean of 5 non-source heads (std 0.011). Top- k consensus = source pair plus next $k-2$ heads from the cross-method consensus ranking ($L_0H_5, L_1H_3, L_2H_1, L_2H_2$). Layer-resolved values in Fig. 6.

Condition	Loose acc.	$\Delta \text{acc.}$
Baseline	0.84	—
Ablate L_0H_0	0.35	+0.48
Ablate L_1H_2	0.78	+0.06
Norm-matched control (mean of 5)	0.84	+0.00
Ablate $\{L_0H_0, L_1H_2\}$	0.20	+0.64
Ablate top-4 (consensus)	0.14	+0.70
Ablate top-6 (consensus)	0.13	+0.71
Ablate all 36 cross-attn heads	0.02	+0.81

180 a functionally specific one. The two source heads are not equally necessary: L_0H_0 carries most of
 181 the causal load on its own, while L_1H_2 's contribution shows up mainly when L_0H_0 is also removed.
 182 We quantify this in the pair-ablation analysis (Section 4.5, $I = +0.086$, 95% CI $[+0.015, +0.157]$).

183 Layer-resolved ablation (Fig. 6) sharpens the localisation: zeroing all six heads in Layer 0 causes
 184 +31 pp damage to loose accuracy, while zeroing any other entire layer changes accuracy by less than
 185 ± 5 pp. The cross-attention spatial signal is concentrated in Layer 0, and within Layer 0 in L_0H_0 .

186 The ablation effect of L_0H_0 is also relation-asymmetric (Fig. 7): diagonals (*upper_right*, *lower_right*,
 187 *lower_left*) take +46–50 pp damage under L_0H_0 ablation, while the four cardinals take +10 to +20
 188 pp damage. Diagonals require both an x- and a y-axis bias, so each L_0H_0 output position contributes
 189 more relation-relevant information for diagonals than for axis-aligned cardinals. L_1H_2 alone has no
 190 measurable effect ($+3$ pp ± 4) on any relation, consistent with the redundancy-with- L_0H_0 picture.

191 Bootstrap CIs from 1000-replicate prompt resamples on this set place $\Delta \text{Acc}_{\text{loose}}(L_0H_0) = +0.48$
 192 (95% CI $[+0.41, +0.55]$) and $\Delta \text{Acc}_{\text{loose}}(L_1H_2) = +0.06$ (95% CI $[+0.02, +0.10]$); the disparity

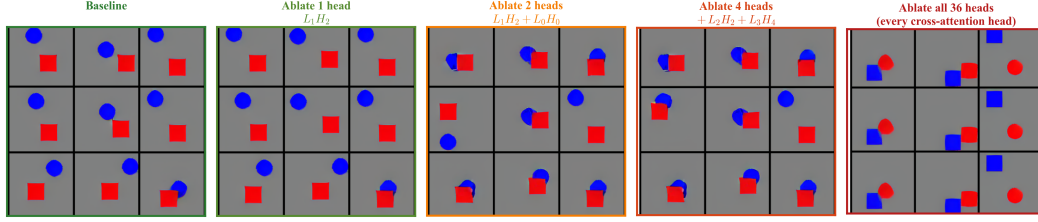


Figure 5: Qualitative cumulative ablation, fixed prompts and seed (epoch 4000, guidance 4.5). Same three relation prompts and same noise seed across all five conditions. *Baseline*: clean two-object layouts. *Ablate L_1H_2 alone*: visually indistinguishable from baseline (consistent with the +6 pp redundant-head finding). *Ablate $\{L_0H_0, L_1H_2\}$* : objects begin merging in the centre as the spatial-routing signal collapses. *Ablate 4 heads*: heavier overlap. *Ablate all 36 cross-attention heads*: layout is fully destroyed but coherent shapes survive (the CFG-induced self-attention object prior).

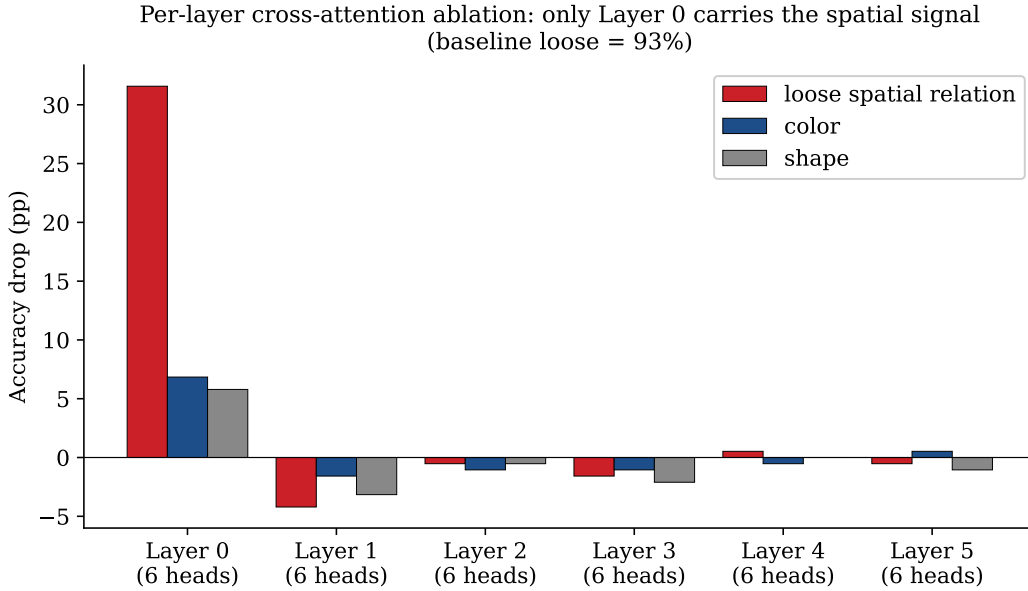


Figure 6: Zero-ablation of every head in a single transformer layer. Layer 0 ablation causes +31 pp damage to loose spatial accuracy; layers 1–5 change accuracy by less than ± 5 pp, indicating the spatial signal originates in layer 0 and propagates through later layers without further read-out via cross-attention. Color and shape accuracies drop only modestly in layer 0 (object identity is largely preserved by self-attention).

193 reflects that, with L_0H_0 intact, L_1H_2 is largely redundant. The mean of the 31 non-source heads is
 194 $\Delta \text{Acc}_{\text{loose}} = +0.003 \pm 0.011$ pp (95% CI excludes L_0H_0 at $> 40\sigma$).

195 On held-out OOD sets the same effect persists, with prompt-bootstrap 95% CIs (Fig. 9): **unseen**
 196 **color** $\times$ **shape** pairs ($n = 12$ prompts, 10 images each) baseline loose 0.80 $\rightarrow$ L_0H_0 -ablated 0.29
 197 ($\Delta = +0.51$, CI $[+0.41, +0.61]$); **synonym-paraphrased relations** ($n = 7$, e.g. “red square is *above*
 198 *and to the left of* blue triangle”) baseline 0.79 $\rightarrow$ ablated 0.60 ($\Delta = +0.19$, CI $[+0.01, +0.33]$).
 199 Both CIs exclude zero. **Harder OOD** variants (distractor, extra-adjective, syntax-fronted; $n=24$):
 200 baseline drops to 0.38–0.42 before any ablation, and L_0H_0 ’s marginal damage shrinks to +7 to +14
 201 pp (17–37% relative vs. 61% on vanilla); the circuit’s effect is largest on the training distribution.
 202 Details in Appendix H.

203 **Redundancy and self-attention prior.** The two source heads project onto overlapping ramp
 204 directions (cosine $\rho \approx 0.7$ between their OV writes) but jointly cover more of the basis than either
 205 alone — the CFG conditioning-dropout setting favours such parallel redundancy. Ablating *all 36*
 206 *cross-attention heads* collapses spatial accuracy to 2% but preserves $\sim 90\%$ colour and $\sim 30\%$

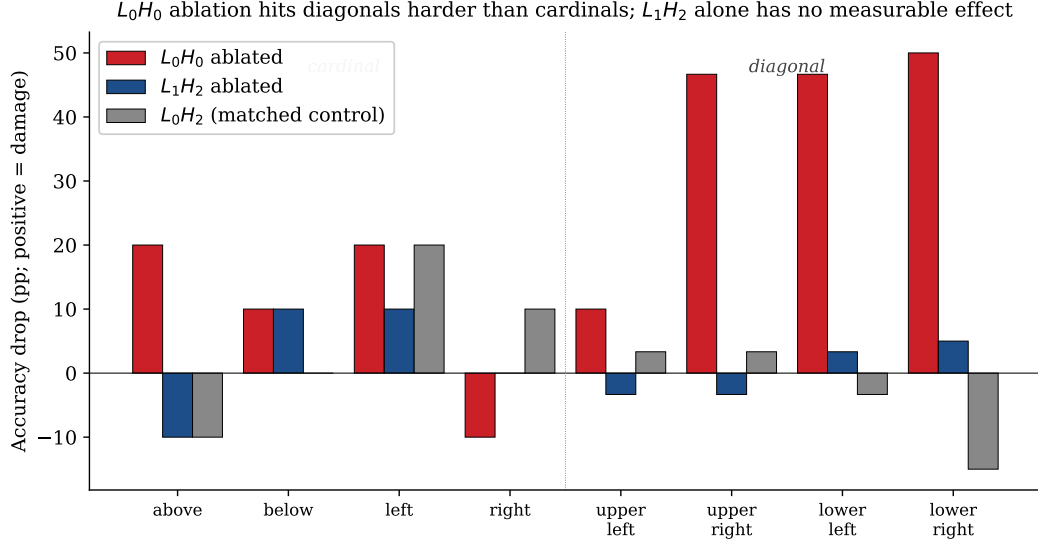


Figure 7: Per-relation ablation effect (positive = damage). L_0H_0 ablation hits diagonals (right four) much harder than cardinals (left four), consistent with diagonals needing two-axis biases. L_1H_2 alone has no measurable effect on any relation. The matched-norm control L_0H_2 shows essentially zero effect.

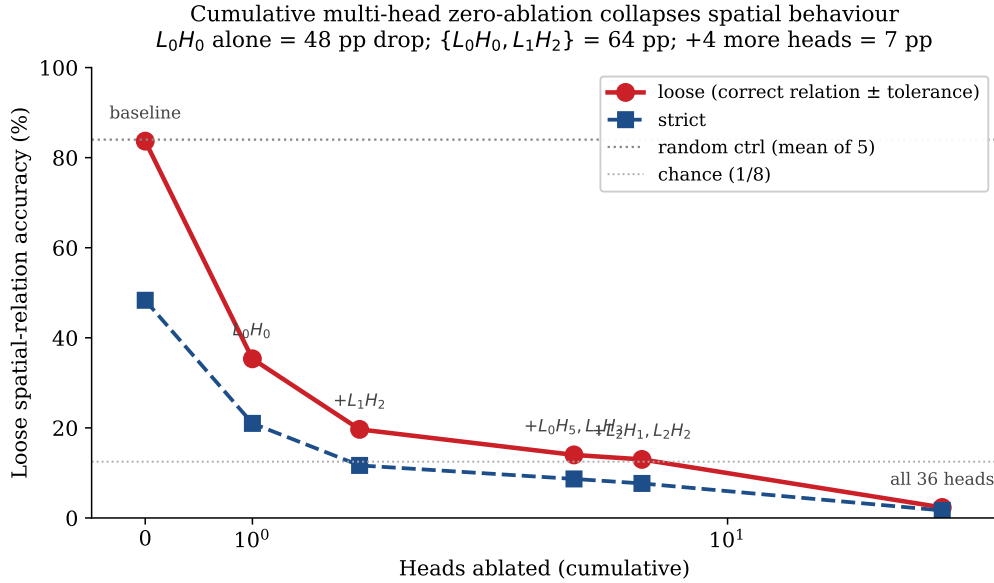


Figure 8: Cumulative multi-head ablation on the 30-prompt training-template set used by pair-ablation: baseline 84% $\rightarrow$ 35% on L_0H_0 alone, $\rightarrow$ 20% when both source heads are zeroed, $\rightarrow$ 13% at top-6, and $\rightarrow$ 2% at all-36. Random control (mean of 5) sits within 0.3 pp of baseline.

207 shape (Table 4); the surviving structure comes from a self-attention object prior trained during
 208 CFG-unconditional steps. We treat this as a secondary observation; full discussion is in Appendix G.

209 4.5 Downstream-reader discovery and causal verification

210 QK-OV composition alone does not cleanly identify a single downstream reader: the top-5 mid-
 211 layer heads have clustered Frobenius scores (18.2–11.5). The four-method consensus (QK-OV,
 212 path-patching, activation correlation, spatial-ramp projection; Appendix D, Table 8) yields top-5

Out-of-distribution generalisation: L_0H_0 ablation effect with 95% prompt-bootstrap CIs

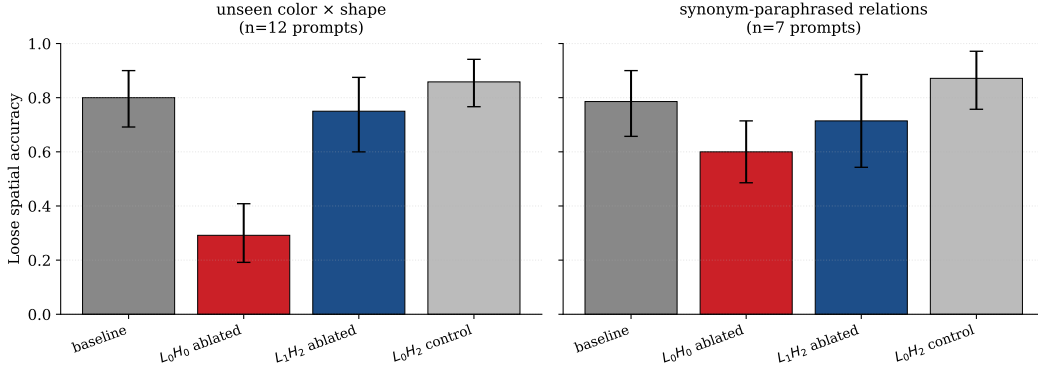


Figure 9: Held-out OOD: loose spatial accuracy with 2000-replicate prompt-bootstrap 95% CIs. L_0H_0 ablation (red) damages both sets significantly; L_1H_2 ablation (blue) is within bootstrap noise of baseline; norm-matched control L_0H_2 (light grey) shows no damage.

Table 5: Pair-ablation interaction scores at the final checkpoint, source = L_0H_0 . Loose-spatial accuracy. $\Delta\text{Acc}(L_0H_0) = +0.483$ for all rows (same source-only condition). Interaction $I = \Delta\text{Acc}(\text{pair}) - \Delta\text{Acc}(L_0H_0) - \Delta\text{Acc}(\text{cand})$; positive = super-additive. Bootstrap 95% CIs from 1000 prompt-resamples.

h	$\Delta(h)$	$\Delta(\text{pair})$	I	95% CI
L_1H_2	+0.060	+0.640	+0.086	[+0.015, +0.157]
L_0H_5	-0.007	+0.480	+0.012	[-0.052, +0.080]
L_1H_3	+0.003	+0.500	+0.008	[-0.026, +0.043]
L_2H_1	+0.027	+0.500	-0.013	[-0.035, +0.013]
L_2H_2	+0.003	+0.487	-0.002	[-0.039, +0.033]

213 $\{L_1H_2, L_0H_5, L_1H_3, L_2H_1, L_2H_2\}$; L_2H_3 (rank-1 by QK-OV) falls to consensus rank 21.5 once
 214 the other signals are included.

215 **Pair-ablation.** For each consensus-top-5 candidate we run the four-condition pair-ablation (baseline,
 216 L_0H_0 only, h only, $\{L_0H_0, h\}$ together) on a 30-prompt held-out set ($n_{\text{img}}=10$). The result (Table 5)
 217 is single-headed: only $\{L_0H_0, L_1H_2\}$ has a super-additive interaction beyond resampling noise
 218 ($I = +0.086$, prompt-bootstrap 95% CI [+0.015, +0.157], 2000 resamples, $n_{\text{prompt}}=23$). The four
 219 other candidates have $|I| \leq 0.013$ with CIs containing zero.

220 **Activation patching.** Patching L_0H_0 's clean head-output into a relation-flipped corrupt forward
 221 pass recovers $\sim 83\%$ of the loose-accuracy gap (clean 0.88 $\rightarrow$ corrupt 0.00 $\rightarrow$ patched 0.71, $n=8$;
 222 Fig. 10) — a converse causal test indicating that L_0H_0 's output contains information sufficient
 223 to restore much of the relation-dependent layout. Patching L_1H_2 recovers 0% despite its positive
 224 pair-ablation interaction; we attribute this to a combination of redundancy with L_0H_0 and patch-site
 225 sensitivity (full discussion in Appendix E).

226 The spatial-only test thus upper-bounds non-source cross-attention contributions at ~ 0 super-additive
 227 interaction. Spatial accuracy is, however, blind to colour/shape binding: a downstream stage that uses
 228 L_0H_0 's output to bind object identity to position would be invisible at this metric. We re-run the same
 229 intervention on colour and shape metrics, where additional structure becomes visible (Section 4.6).
 230 The metric-resolved pair-ablation table (showing L_0H_0 alone already damages colour by 15 pp and
 231 shape by 26 pp, but consensus candidates add no further binding damage at this resolution) is in
 232 Appendix F, Table 9. The held-out generalisation pattern persists for the source pair ($I = +0.084$,
 233 CI [+0.05, +0.12]) on the unseen color $\times$ shape set.

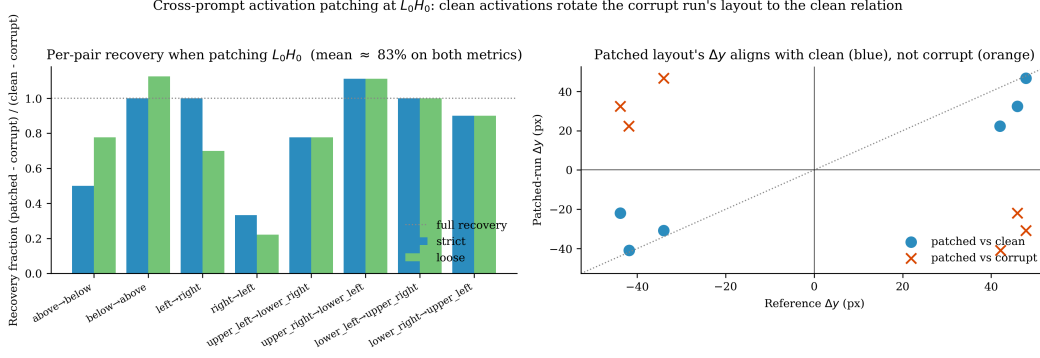


Figure 10: Cross-prompt activation patching at L_0H_0 . *Left*: per-relation-flip recovery fraction; L_0H_0 patching recovers $\sim 83\%$. *Right*: patched layout's Δy aligns with clean (diagonal), not corrupt — L_0H_0 's head-output causally controls placement direction.

Table 6: Layer-pair interaction $I(\text{layer } L \mid \text{layer } 0)$ in pp; positive = super-additive. Same 19-prompt training-template set.

L	I_{spatial}	I_{color}	I_{shape}
1	+14.2	-7.4	-7.9
2	+7.4	+16.8	+12.6
3	-9.5	-1.6	-8.9
4	+4.7	+7.4	-3.2
5	-2.1	+0.5	-1.6

234 4.6 Downstream binding stage: Layer 2 super-additive on object identity

235 The pair-ablation test in Table 5 measures super-additivity on *spatial* accuracy. Spatial accuracy
 236 treats colour and shape as nuisance variables: an image with a red triangle in the upper-right and
 237 a blue square in the lower-left scores correctly for “red triangle is above blue square” regardless
 238 of whether the colours/shapes are right. So pair-ablation on spatial accuracy is insensitive to any
 239 downstream stage that uses L_0H_0 's output to bind colours and shapes to positions. We re-ran the
 240 same pair-ablation analysis on the colour and shape metrics, and on a layer-resolved variant.

241 **Layer-resolved pair-ablation.** For each layer $L \in \{1, \dots, 5\}$ we zero all six heads of layer 0 and
 242 additionally all six heads of layer L . The interaction $I(\text{layer } L \mid \text{layer } 0) = \Delta(\text{both}) - \Delta(\text{layer } 0) -$
 243 $\Delta(\text{layer } L \text{ alone})$ on three metrics is shown in Fig. 1(b) and Table 6.

244 Layer 2 stands out: zeroing it on top of layer 0 produces a +16.8 pp super-additive damage on colour
 245 and +12.6 pp on shape, while layer 2 zeroed alone has essentially no effect (≤ 1 pp on every metric).
 246 We interpret this as evidence for layer 2 acting downstream of layer 0 on the colour/shape-to-position
 247 assignment: ablating layer 2 alone has little effect, but removing it together with layer 0 further
 248 degrades colour and shape accuracy. We emphasise that the point here is not the exact localisation to
 249 layer 2, but that the effect appears only when colour and shape are scored separately from spatial
 250 accuracy.

251 **Single-head triple ablation.** At single-head resolution, we ran the triple-ablation $I(h \mid$
 252 $\{L_0H_0, L_1H_2\}) = \Delta(\text{src} + h) - \Delta(\text{src}) - \Delta(h \text{ alone})$ for the consensus candidates plus four heads
 253 with the largest single-head colour or shape damage on the held-out set ($L_0H_4, L_0H_1, L_1H_5, L_0H_2$).
 254 Several candidates are super-additive on *binding* accuracy when ablated jointly with the source pair
 255 (Table 7): L_0H_5 on colour (+7.9) and shape (+5.8); L_2H_2 on colour (+10.0); L_2H_1 on colour
 256 (+7.4); L_0H_4 on colour (+6.3). Spatial-resolved interactions are smaller (+2–+7 pp) and roughly
 257 half the binding-resolved interactions, consistent with these heads contributing primarily to object
 258 binding rather than spatial routing.

Table 7: Triple-ablation interaction $I(h \mid \text{src})$ in pp where $\text{src} = \{L_0H_0, L_1H_2\}$. Sorted by mean of colour/shape interactions. The first six rows have super-additive colour interactions $\geq +6$ pp; the bottom two (L_0H_1, L_1H_5) are weaker controls.

Head h	I_{spatial}	I_{color}	I_{shape}
L_0H_5	+6.8	+ 7.9	+ 5.8
L_2H_2	+5.8	+ 10.0	+1.1
L_2H_1	+4.2	+ 7.4	+2.6
L_0H_4	+5.8	+6.3	+3.2
L_1H_3	+5.3	+5.8	+1.1
L_0H_2	+3.7	+1.6	+5.3
L_0H_1	+2.6	+3.2	−0.5
L_1H_5	+2.1	−3.2	+0.0

Why the original pair-ablation missed this. At the single-head level, no layer-2 head individually damages binding by more than 1–3 pp (max $\Delta_{\text{shape}} = 4.7$ pp at L_2H_3). The binding stage is *distributed* across the layer; single-head pair-ablation was blind to it because no individual layer-2 head carries enough damage to register against bootstrap noise. This is consistent with the Wang et al. [2026] observation that DiTs trained with pretrained T5 (vs. random embeddings) recruit a broader pool of heads — here, that pool is Layer 2. Bar-chart visualisation of the triple-ablation interactions is in Appendix I, Fig. 12.

4.7 Limitations

Single seed: all results are on one training run; we report prompt-bootstrap (evaluation) CIs but not training-noise CIs. **Scale:** the model has 36 heads; frontier DiTs have hundreds. **Distributed binding:** we localise the binding stage to layer 2 as a group, but no single layer-2 head shows > 3 pp damage; finer localisation needs denser combinatorial ablation. **Time-aggregated:** step-resolved ablation across the noise schedule is left for future work.

5 Conclusion

Within this controlled object-relation testbed on a small PixArt-mini DiT, we identify a two-stage cross-attention circuit for spatial-relation binding at head resolution: a source pair $\{L_0H_0, L_1H_2\}$ in layers 0–1 ($I = +0.086$ for the pair; $|I| \leq 0.013$ for all other tested heads), and a distributed object-binding stage at Layer 2 that is strongly super-additive on colour and shape when co-ablated with layer 0, despite little effect alone. Methodologically, the layer-2 stage is invisible when only spatial accuracy is scored; in multi-attribute generative systems, ablation studies that score a single metric can miss downstream stages that act on the others, and similar care is likely needed in LM circuit work spanning factual, stylistic, and structural axes. Open directions include multi-seed replication, denser per-head colour/shape ablation, step-resolved ablation, and scaling to larger DiTs.

References

- Samyadeep Basu, Nanxuan Zhao, Vlad Morariu, Soheil Feizi, and Dinesh Manocha. Localizing and editing knowledge in text-to-image generative models. In *International Conference on Learning Representations (ICLR)*, 2024.
- Hila Chefer, Yuval Alaluf, Yael Vinker, Lior Wolf, and Daniel Cohen-Or. Attend-and-excite: attention-based semantic guidance for text-to-image diffusion models. In *ACM Transactions on Graphics (SIGGRAPH)*, 2023.
- Junsong Chen, Jincheng Yu, Chongjian Ge, Liewen Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu, and Zhenguo Li. PixArt- α : Fast training of diffusion transformer for photorealistic text-to-image synthesis. In *International Conference on Learning Representations (ICLR)*, 2024.

293 Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-
294 Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in*
295 *Neural Information Processing Systems (NeurIPS)*, 2023.

296 Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda
297 Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac
298 Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse,
299 Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A
300 mathematical framework for transformer circuits. Transformer Circuits Thread, 2021.

301 Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural
302 networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

303 Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. arXiv:2404.15255,
304 2024.

305 Alec Helbling et al. ConceptAttention: Diffusion transformers learn highly interpretable features. In
306 *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.04320.

307 Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-
308 to-prompt image editing with cross-attention control. In *International Conference on Learning*
309 *Representations (ICLR)*, 2023.

310 Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS Workshop on Deep*
311 *Generative Models and Downstream Applications*, 2022.

312 Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in*
313 *Neural Information Processing Systems (NeurIPS)*, 2020.

314 Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual
315 associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

316 Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for
317 grokking via mechanistic interpretability. In *International Conference on Learning Representations*
318 *(ICLR)*, 2023.

319 Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan,
320 Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads.
321 Transformer Circuits Thread, 2022.

322 Hadas Orgad, Bahjat Kawar, and Yonatan Belinkov. Editing implicit assumptions in text-to-image
323 diffusion models. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

324 William Peebles and Saining Xie. Scalable diffusion models with transformers. In *IEEE/CVF*
325 *International Conference on Computer Vision (ICCV)*, 2023.

326 Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: General-
327 ization beyond overfitting on small algorithmic datasets. arXiv:2201.02177, 2022.

328 Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi
329 Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text
330 transformer. *Journal of Machine Learning Research (JMLR)*, 2020.

331 Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-
332 resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer*
333 *Vision and Pattern Recognition (CVPR)*, 2022.

334 Raphael Tang, Linqing Liu, Akshat Pandey, Zhiying Jiang, Gefei Yang, Karun Kumar, Pontus
335 Stenetorp, Jimmy Lin, and Ferhan Ture. What the DAAM: Interpreting stable diffusion using cross
336 attention. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

337 Binxu Wang, Jingjie Fan, and Xinyi Pan. Circuit mechanisms for spatial relation generation in
338 diffusion transformers. arXiv:2601.06338, Kempner Institute, Harvard University, 2026.

- 339 Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Inter-
340 pretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International*
341 *Conference on Learning Representations (ICLR)*, 2023.
- 342 Arman Zarei, Samyadeep Basu, et al. Localizing knowledge in diffusion transformers. In *Advances*
343 *in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2505.18832.
- 344 Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models:
345 metrics and methods, 2024.

346 A Experimental setup details

347 **Architecture.** Cross-attention has the conventional form

$$\text{Attn} = \text{softmax}\left(\frac{XW_Q(EW_K)^\top}{\sqrt{d_{\text{head}}}}\right)EW_V, \quad (2)$$

348 with $X \in \mathbb{R}^{256 \times 384}$ (spatial latent tokens on a 16×16 grid), $E \in \mathbb{R}^{L_{\text{tok}} \times 4096}$ (T5 embeddings,
349 $L_{\text{tok}} \approx 15$), $d=384$, $d_{\text{head}}=64$, $H=6$ heads/layer, $L=6$ layers. Training: 4000 epochs with EMA,
350 CFG conditioning-dropout rate 0.1.

351 **Held-out sets.** (i) *Unseen color $\times$ shape*: $n=12$ prompts using colour/shape combinations not seen
352 during training. (ii) *Synonym-paraphrased relations*: $n=7$ prompts substituting synonyms (*on top of*,
353 *underneath*, *above* and *to the left of*). (iii) Harder OOD: see Appendix H.

354 **OpenCV evaluator.** For each 128×128 image, we threshold to a binary mask, find external
355 contours via `findContours(RETR_EXTERNAL)`, classify each contour by mean RGB inside the
356 contour (within margin 25 of canonical red/blue) and by polygon-segment count from `approxPolyDP`
357 ($3 \rightarrow \text{triangle}$, $4 \rightarrow \text{square}$, $\geq 7 \rightarrow \text{circle}$). For an image to score, both objects must be detected
358 with the right colour and shape and their centroids’ relative offset ($\Delta x, \Delta y$) must satisfy the relation.
359 *Strict*: $|\Delta x|, |\Delta y| \geq 5$ pixels. *Loose*: ≥ 8 pixels with the right sign for the relation (e.g. *upper_right*
360 requires $\Delta x \geq 8$ and $\Delta y \leq -8$). Strict additionally requires no spatial overlap.

361 **Inference settings.** DPM-Solver multi-step scheduler at 14 inference steps, CFG scale 4.5 (match-
362 ing training), 10 images per prompt per condition. Pilot single-step ablations produced < 2 pp change,
363 so we report all-step ablation throughout.

364 B Candidate-head discovery details

365 **Spatial-ramp basis.** The basis consists of $D=8$ relation directions $\mathcal{D} = \{\text{above, below, left,}$
366 $\text{right, upper_left, upper_right, lower_left, lower_right}\}$. For each r with unit vector $(d_y^{(r)}, d_x^{(r)}) \in$
367 $\{-1, 0, +1\}^2$, the ramp $\mathbf{R}_r \in \mathbb{R}^{64}$ over the 8×8 patch grid is $[\mathbf{R}_r]_{(i,j)} = d_y^{(r)} \tilde{y}_i + d_x^{(r)} \tilde{x}_j$ with
368 $\tilde{y}_i, \tilde{x}_j \in \text{linspace}(-1, 1, 8)$, mean-centred. The latent has 16×16 patches but the cross-attention
369 output factorises within 8×8 super-tiles, so we operate at 8×8 .

370 **Effect-vector decomposition.** For each prompt embedding $\mathbf{e}_i \in \mathbb{R}^{4096}$ we run an additive-effects
371 regression over factors $\mathcal{F} = \{\text{rel, color}_1, \text{color}_2, \text{shape}_1, \text{shape}_2, \text{interactions}\}$:

$$\mathbf{e}_i = \boldsymbol{\mu} + \sum_{f \in \mathcal{F}} \alpha_{f,i} \mathbf{v}_f + \boldsymbol{\epsilon}_i. \quad (3)$$

372 Pushing $\mathbf{v}_{\text{rel}}$ through the caption projection gives $\mathbf{p}_{\text{rel}} = W_{\text{cap}} \mathbf{v}_{\text{rel}} \in \mathbb{R}^{384}$. The variance partition
373 reports $R_f^2 = 1 - \text{SS}_{\text{resid, w/o } f} / \text{SS}_{\text{total}}$.

374 **Permutation null.** 200 shuffled-relation refits give a null distribution that peaks at $|p|=8.3$ with
375 mean 4.1 ± 1.6 . $L_0 H_0$ ’s observed $|p|=47.7$ is a $\sim 27\sigma$ departure ($p < 10^{-4}$). A shape-identity
376 negative control gives $|p| < 3$ for every shortlisted head, ruling out a generic-high-variance artefact.

377 C Causal ablation details

378 **Zero-head ablation.** With the per-head attention output tensor of shape $(B, H, P, d_{\text{head}})$ where
379 B is the CFG-doubled batch ($B=2$), $H=6$ heads, $P=256$ image patches, $d_{\text{head}}=64$, we set `tensor[:,`
380 `h_target, :, :] = 0` at every denoising step and on both CFG branches. The per-head output is zeroed
381 *before* concatenation and projection through W_O .

382 **Norm-matched random control.** Five non-shortlist heads of comparable OV Frobenius norm.
383 Mean ablation effect -0.003 ± 0.011 pp (95% CI excludes $L_0 H_0$ at $> 40\sigma$).

D Downstream-reader search details

Method definitions.

- **M1 (offline OV-QK alignment):** rank by $|\cos|$ between candidate’s OV write and the spatial-ramp basis.
- **M2 (path-patching recovery loss):** from a clean / relation-flipped pair, patch L_0H_0 from clean (restoring attention) and additionally re-zero h_d ; the recovery loss measures h_d ’s on-path causal contribution.
- **M3 (in-vivo activation correlation):** capture every head’s per-step head-output for 30 prompts, compute per-prompt Pearson correlation between L_0H_0 and h_d , average across prompts.
- **M4 (per-head spatial-ramp projection):** per-position ℓ_2 norm of h_d ’s head-output (averaged over time and head_dim) correlated with the canonical 8×8 ramp R_r for the prompt’s relation; report mean $|\cos|$ across prompts.

For each method we rank the 35 non- L_0H_0 heads. Path-patching does not sweep L_0 , so L_0 entries get the column median. Consensus is the unweighted mean rank.

Table 8: Cross-method consensus ranking. Each cell is the head’s rank in that method (1 = best); consensus is the mean rank.

Cand.	M1	M2	M3	M4	Cons.
L_1H_2	2	11	7	2	5.50
L_0H_5	5	11	3	6	6.25
L_1H_3	6	1	10	17	8.50
L_2H_1	8	3	12	14	9.25
L_2H_2	14	3	20	1	9.50
L_2H_3	19	32	23	12	21.50
L_3H_1	15	3	16	30	16.00
L_4H_2	26	3	14	21	16.00

Activation-patching protocol. Patches are applied at the per-head attention output (post-OV, pre-projection) of h_d , on the conditional half of the CFG batch only, at all 14 denoising steps. Eight relation-flip pairs: above↔below, left↔right, and the four diagonals.

E Reconciling the patching null with pair-ablation

L_1H_2 has a positive pair-ablation interaction ($I = +0.086$, CI excludes zero) but a 0% activation-patching recovery. Three explanations are consistent with this asymmetry:

- **Redundancy.** L_1H_2 writes a partially-redundant copy of L_0H_0 ’s output; in the corrupt forward pass L_0H_0 already injects the flipped pattern into the residual, so patching L_1H_2 from clean adds the same direction without overriding it. Joint ablation eliminates both copies at once.
- **Interaction-only contribution.** If L_1H_2 contributes only when the source-pair output exceeds a threshold, single-head patching cannot trigger it. The pair-ablation interaction is positive because L_1H_2 ’s contribution becomes detectable only when L_0H_0 is removed.
- **Patching-site mismatch.** We patch at the per-head attention output (post-OV, pre-projection); a more sensitive site might be the residual stream after the layer’s output projection.

Discriminating among these requires denser patching experiments (residual-stream patching, scaled patching, partial-replacement patching), left for future work.

Table 9: Metric-resolved pair-ablation. Δ = baseline – ablated; positive = damage. 30-prompt held-out set.

Condition	Δ_{rel}	Δ_{col}	Δ_{sh}	Δ_{bind}
L_0H_0 alone	+0.483	+0.147	+0.263	+0.273
$\{L_0H_0, L_1H_2\}$	+0.640	+0.180	+0.270	+0.310
$\{L_0H_0, L_0H_5\}$	+0.480	+0.190	+0.267	+0.293
$\{L_0H_0, L_1H_3\}$	+0.500	+0.210	+0.283	+0.297
$\{L_0H_0, L_2H_1\}$	+0.500	+0.173	+0.283	+0.300
$\{L_0H_0, L_2H_2\}$	+0.487	+0.140	+0.260	+0.267

417 F Metric-resolved pair-ablation table

418 G Self-attention prior under full cross-attention ablation

419 Ablating all 36 cross-attention heads collapses spatial accuracy to 2% but preserves $\sim 90\%$ colour
 420 and $\sim 30\%$ shape (Table 4, last row). This surviving generation comes from self-attention, trained
 421 during CFG-unconditional steps ($p(\text{img})$, no text) to produce coherent shaped objects without any
 422 text conditioning. Relational information enters only via cross-attention, so removing it destroys
 423 the spatial signal but leaves the self-attention object prior intact. The two source heads project onto
 424 overlapping ramp directions (cosine $\rho \approx 0.7$ between their OV writes) but jointly cover more of the
 425 basis than either alone — the parallel redundancy is consistent with CFG dropout favouring such
 426 backup configurations.

427 H Harder-OOD prompt variants

428 We built three harder OOD variants and re-encoded 24 prompts via T5-XXL: *distractor* (filler clause
 429 appended: “red square is above blue circle, in a quiet scene”), *extra adjectives* (“the small red
 430 square is above the large blue circle”), and *syntax-fronted* (“Above the blue circle is the red square”).
 431 The vanilla effect (+50 pp damage, baseline 0.83) reproduces the in-distribution result. On all
 432 three harder variants the model’s baseline drops to 0.38–0.42 before any ablation. L_0H_0 ablation
 433 produces additional damage on every variant (+7 to +14 pp), with relative drops $|\Delta|/\text{baseline}$ of
 434 17% (distractor), 37% (extra adjectives), and 22% (syntax-fronted), versus 61% on vanilla. The
 435 circuit is not invariant to prompt syntax; its absolute effect is largest on the training distribution. This
 436 bounds the generalisation claim and identifies prompt-syntax robustness as a separate axis.

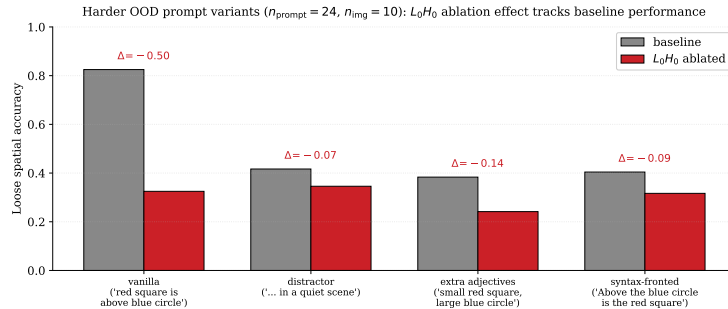


Figure 11: Harder OOD prompt variants. Vanilla template: L_0H_0 ablation causes +50 pp damage. The three harder variants degrade the model’s baseline accuracy substantially (from 0.83 to 0.38–0.42); L_0H_0 ’s marginal contribution becomes correspondingly smaller in absolute terms (+7 to +14 pp).

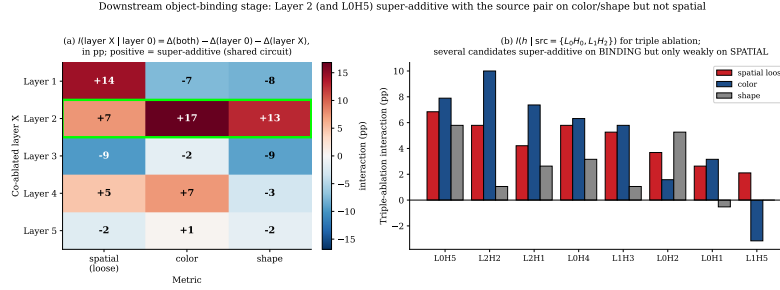


Figure 12: (a) Layer-pair interaction with layer 0; layer 2 (lime) shows positive interaction on colour and shape, but only a smaller interaction on spatial accuracy. (b) Triple-ablation interaction $I(h | \{L_0H_0, L_1H_1\})$. Spatial interactions (red bars) are systematically smaller than the corresponding colour interactions (blue bars).

I Triple-ablation bar chart

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 state the four contributions (two-stage circuit, causal necessity, sharp emergence, metric-resolved discovery), explicitly bound the scope to a small PixArt-mini DiT in a controlled object-relation testbed, and disclose that all causal and emergence results come from a single training run with prompt-bootstrap CIs (not training-seed CIs). Numerical claims in the abstract match the values reported in Sections 4.4–4.6.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 4.7 (“Limitations”) explicitly discusses single-seed scope, model scale (36 heads vs. hundreds in frontier DiTs), distributed binding (no individual layer-2 head shows > 3 pp damage so finer localisation needs denser combinatorial ablation), and time-aggregated ablation. The single-seed limitation is also surfaced earlier at the end of Section 1.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.

- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is empirical; it contains no formal theorems or proofs. Definitions used (projection magnitude, super-additive interaction I , variance partition R^2) are stated where introduced in Section 3 and Appendices B–C.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 specifies the architecture, training recipe (4000 epochs, EMA, CFG dropout 0.1), evaluation protocol (post-hoc OpenCV detection, prompt-bootstrap CIs), candidate-head discovery score, ablation site (per-head attention output, post-OV pre-concat, both CFG branches, every denoising step), and downstream-reader search (four methods, consensus rank). Appendix A documents tensor shapes, scheduler/CFG settings, evaluator thresholds (strict vs. loose), and held-out sets. Anonymized code is provided as supplemental material (see Q5).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: An anonymized version of the analysis code (head discovery, zero/pair/triple ablation, activation patching, checkpoint emergence sweeps, figure-generation scripts) is included as supplemental material with a README documenting environment setup and the exact commands to reproduce each main-text figure and table. The synthetic prompt set and trained model checkpoints are released alongside the code (see Q13).

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.

- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 (“Methods”) specifies architecture ($L = 6$, $H = 6$, $d = 384$, $d_{\text{head}} = 64$), conditioning (frozen T5-XXL with learned $\mathbb{R}^{4096} \rightarrow \mathbb{R}^{384}$ caption projection), training (4000 epochs, batch 256, EMA, CFG dropout 0.1, 128×128 images), evaluation (250 sampling steps, guidance 4.5, 10 images/prompt, OpenCV centroid+RGB+polygon detection), and the eleven training checkpoints used for the emergence analysis. Held-out sets and evaluator thresholds (strict vs. loose pixel margins) are detailed in Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All accuracy and interaction estimates report 1000- or 2000-replicate prompt-bootstrap 95% confidence intervals (resampling held-out evaluation prompts with replacement); the bootstrap is the only randomness source captured, and the paper explicitly notes this in Section 1 (“Scope”). Head-discovery significance uses a 200-permutation null over relation labels with a shape-identity negative control (Section 4.1). Reviewer-relevant null results (e.g. $L_1 H_2$ patching) include the CI explicitly. CIs are asymmetric where appropriate; the bars in figures and the bracketed values in tables are the bootstrap CIs.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).

- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments were run on a single NVIDIA A100 (80 GB) on an academic compute cluster. PixArt-mini training (4000 epochs, batch 256, 128×128) takes approximately 15 GPU-hours; the full set of analysis sweeps reported in the paper (zero/pair/triple ablations across 36 heads, layer-pair sweeps, cross-prompt activation patching, the eleven-checkpoint emergence re-evaluation, and held-out OOD evaluation) collectively account for approximately 80 GPU-hours. Total compute, including preliminary scans and discarded ablation conditions, is estimated at ~ 120 GPU-hours. Inference VRAM peaks at ~ 3 GB.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work uses synthetic prompts and a small academic-scale model trained on synthetic two-object scenes; it involves no human subjects, no scraped data, no personally identifiable information, and no deployment of generative content. The authors have reviewed the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: This is foundational mechanistic-interpretability research on a small, synthetic-data DiT; its primary societal value is positive (interpretability and circuit-level transparency of generative models, which supports auditing, safety, and debugging). It does not improve generation quality, scale, or controllability of any production system, and the model studied is too small for direct misuse. Negative downstream uses we can anticipate (e.g. targeted concept-removal attacks on production T2I models) would require substantially extending the methods to frontier-scale DiTs and are not enabled by the present work in isolation.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.

- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The released model is a small PixArt-mini variant trained from scratch on synthetic two-object scenes with ~ 1000 prompts of the form “[colour] [shape] is [relation] [colour] [shape]”. It cannot generate photorealistic, person-related, or otherwise sensitive imagery, and poses no realistic misuse risk. The released dataset is fully synthetic and contains no scraped, copyrighted, or PII-bearing content.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The paper builds on PixArt- α [Chen et al., 2024] (AGPL-3.0 architecture; we re-implement the “mini” variant from scratch), the T5-XXL text encoder [Raffel et al., 2020] (Apache-2.0), OpenCV (Apache-2.0) for the post-hoc evaluator, and PyTorch (BSD-3); each is cited at first use. No third-party datasets are used (training prompts are generated programmatically). The released code uses the MIT License and the released checkpoints/dataset are released under CC-BY-4.0; both are documented in the supplementary README.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: Three new assets accompany the submission, all anonymized for review: (i) the analysis-code repository (head discovery, ablation, patching, emergence-sweep, and figure scripts) with a README documenting environment, commands, and exact figure/table mapping; (ii) the trained PixArt-mini checkpoint at epoch 4000 plus the eleven intermediate checkpoints used in the emergence analysis ({100, 250, 500, 600, 700, 750, 800, 900, 1000, 2000, 4000}); (iii) the synthetic prompt set (~1000 prompts, 2 colours $\times$ 3 shapes $\times$ 8 relations) with the cached T5-XXL embeddings. Licenses (MIT for code, CC-BY-4.0 for model and data) and intended-use notes are documented in the package README.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs (Claude) were used as a coding assistant to help implement individual analysis scripts (e.g. ablation hooks, plotting code) and to assist with manuscript editing. They were not part of the core methodology: the experimental design, choice of methods (variance partition, QK–OV alignment, four-method consensus, pair/triple ablation, activation patching, checkpoint emergence sweep), interpretation of results, and conclusions are the authors' own.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.